UNIVERSITATEA DIN BUCUREȘTI
FACULTATEA DE MATEMATICĂ ȘI INFORMATICĂ

TUTORIAT
CALCUL NUMERIC
II

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Factorizarea LU (2)

Exemplu:

$$Sa\ se\ scrie\ matricea\ A = \begin{pmatrix} 1 & 1 & -4 & 0 \\ 1 & 3 & -1 & 1 \\ 0 & 4 & 7 & 5 \\ 3 & 5 & -7 & 10 \end{pmatrix} sub\ forma\ A = L * U, unde$$

$$L\ respectiv\ U\ sunt\ matrice\ inferior\ triunghiulara\ respectiv\ superior$$

$$triunghiulara.$$

Rezolvare:

$$Dupa\ ce\ verificam\ daca\ poate\ fi\ aplicata\ factorizarea\ (vezi\ curs):$$

$$\begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 & 0 & 0 \\ l_{10} & l_{11} & 0 & 0 \\ l_{20} & l_{21} & l_{22} & 0 \\ l_{30} & l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ 0 & u_{11} & u_{12} & u_{13} \\ 0 & 0 & u_{22} & u_{23} \\ 0 & 0 & 0 & u_{33} \end{pmatrix}$$

$$\begin{pmatrix} a_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 \\ L_{10} & L' \end{pmatrix} \begin{pmatrix} u_{00} & U_{01} \\ 0 & U' \end{pmatrix}$$

$$Obtinem\ sistemul:$$

$$\begin{cases} a_{00} = l_{00}u_{00} & (1) \\ A_{01} = l_{00}U_{01} & (2) \\ A_{10} = u_{00}L_{10} & (3) \\ A_{11} = L_{10}U_{01} + L'U' & (4) \end{cases}$$

$$Din\ ecuatia\ (1)\ alegem\ l_{00} = 1 \Longrightarrow a_{00} = u_{00} \Longrightarrow u_{00} = 1$$

$$Din\ ecuatia\ (2): A_{01} = U_{01} \Longrightarrow \begin{cases} u_{01} = 1 \\ u_{02} = -4 \\ u_{03} = 0 \end{cases}$$

$$Din\ ecuatia\ (3): A_{10} = L_{10} \Longrightarrow \begin{cases} l_{10} = 1 \\ l_{20} = 0 \\ l_{30} = 3 \end{cases}$$

$Vrem\ sa\ aflam\ L'U'din\ ecuatia\ (4)si\ putem\ calcula\ L_{10}U_{01}\ (aflate\ anterior), iar$

$L'U' = A_{11} - L_{10}U_{01}.$

$$L_{10}U_{01} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & -4 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -4 & 0 \\ 0 & 0 & 0 \\ 3 & -12 & 0 \end{pmatrix}.$$

$$L'U' = \begin{pmatrix} 3 & -1 & 1 \\ 4 & 7 & 5 \\ 5 & -7 & 10 \end{pmatrix} - \begin{pmatrix} 1 & -4 & 0 \\ 0 & 0 & 0 \\ 3 & -12 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix} = A'$$

$$A' = L'U'$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

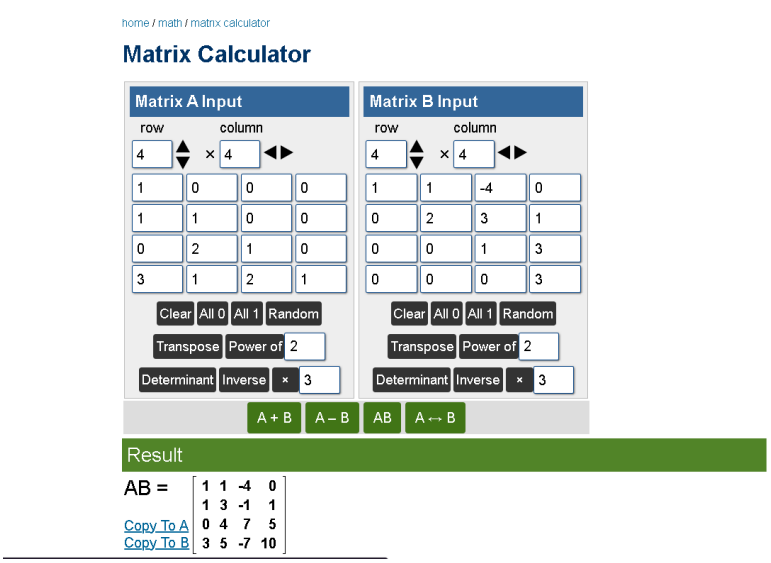
$In\ continuare\ aplicam\ acelasi\ algortim\ de\ factorizare\ LU\ pentru\ matricea\ A'. Totusi,$

$$observam\ faptul\ ca\ A' = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix}\ (vezi\ in\ Tutoriatul\ 1)$$

$Obtinem$:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 3 & 1 & 2 & 1 \end{pmatrix} \qquad U = \begin{pmatrix} 1 & 1 & -4 & 0 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

$Verificare$:



Factorizarea UL

Exemplu:

$$Sa\ se\ scrie\ matricea\ A = \begin{pmatrix} 1 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix}\ sub\ forma\ A = U * L, unde$$

$$U\ respectiv\ L\ sunt\ matrice\ superior\ triunghiulara\ respectiv\ inferior$$

$$triunghiulara.$$

Rezolvare:

$$Dupa\ ce\ verificam\ daca\ poate\ fi\ aplicata\ factorizarea\ (vezi\ curs):$$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix} = \begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} u_{00} & u_{01} & u_{02} \\ 0 & u_{11} & u_{12} \\ 0 & 0 & u_{22} \end{pmatrix} \begin{pmatrix} l_{00} & 0 & 0 \\ l_{10} & l_{11} & 0 \\ l_{20} & l_{21} & l_{22} \end{pmatrix}$$

$$Rescriem\ matricele\ pe\ blocuri:$$

$$\begin{pmatrix} A_{11} & A_{02} \\ A_{20} & a_{22} \end{pmatrix} = \begin{pmatrix} U_{11} & U_{02} \\ 0 & u_{22} \end{pmatrix} \begin{pmatrix} L_{11} & 0 \\ L_{20} & l_{22} \end{pmatrix}$$

$$Obtinem\ sistemul:$$

$$\begin{cases} A_{11} = U_{11}L_{11} + U_{02}L_{20} \\ A_{02} = l_{22}U_{02} \\ A_{20} = u_{22}L_{20} \\ a_{22} = u_{22}l_{22} \end{cases}$$

$$Obtinem\ o\ infinitate\ de\ produse\ UL = A, deci\ alegem\ u_{22} = 1$$

$$Sistemul\ devine: \begin{cases} A_{11} = U_{11}L_{11} + U_{02}L_{20} & (4) \\ A_{02} = l_{22}U_{02} & (3) \\ A_{20} = L_{20} & (2) \\ a_{22} = l_{22} & (1) \end{cases}$$

$Din\ ecuatia\ (1) \Longrightarrow l_{22} = 10$

$Din\ ecuatia\ (2): A_{20} = L_{20} \Longrightarrow \begin{cases} l_{20} = 2 \\ l_{21} = 5 \end{cases}$

$Din\ ecuatia\ (3): A_{02} = 10U_{02} \Longrightarrow \begin{cases} a_{02} = 10u_{02} \\ a_{12} = 10u_{12} \end{cases} \Longrightarrow \begin{cases} 1 = 10u_{02} \\ 5 = 10u_{12} \end{cases} \Longrightarrow$

$$\Longrightarrow \begin{cases} u_{02} = \dfrac{1}{10} \\ u_{12} = \dfrac{5}{10} = \dfrac{1}{2} \end{cases}$$

$Vrem\ sa\ aflam\ U_{11}L_{11}\ si\ putem\ calcula\ U_{02}L_{20}\ (aflate\ anterior), iar$

$U_{11}L_{11} = A_{11} - U_{02}L_{20}$

$$U_{02}L_{20} = \begin{pmatrix} \frac{1}{10} \\ \frac{1}{2} \end{pmatrix} (2 \quad 5) = \begin{pmatrix} \frac{1}{10} & \frac{5}{10} \\ \frac{1}{1} & \frac{5}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{1}{2} \\ 1 & \frac{5}{2} \end{pmatrix}$$

$$U_{11}L_{11} = A_{11} - U_{02}L_{20} = \begin{pmatrix} 2 & 3 \\ 4 & 7 \end{pmatrix} - \begin{pmatrix} \frac{1}{5} & \frac{1}{2} \\ 1 & \frac{5}{2} \end{pmatrix} = \begin{pmatrix} \frac{9}{5} & \frac{5}{2} \\ 3 & \frac{9}{2} \end{pmatrix}$$

Cine e $U_{11}L_{11}$?

$$U_{11}L_{11} = \begin{pmatrix} u_{00} & u_{01} \\ 0 & u_{11} \end{pmatrix} \begin{pmatrix} l_{00} & 0 \\ l_{10} & l_{11} \end{pmatrix} = \begin{pmatrix} u_{00}l_{00} + u_{01}l_{10} & l_{11}u_{01} \\ u_{11}l_{10} & u_{11}l_{11} \end{pmatrix};$$

$$dar\ U_{11}L_{11} = \begin{pmatrix} \frac{9}{5} & \frac{5}{2} \\ 3 & \frac{9}{2} \end{pmatrix} \Longrightarrow \begin{cases} u_{00}l_{00} + u_{01}l_{10} = \frac{9}{5} & (8) \\ l_{11}u_{01} = \frac{5}{2} & (7) \\ u_{11}l_{10} = 3 & (6) \\ u_{11}l_{11} = \frac{9}{2} & (5) \end{cases}$$

$Alegem\ u_{11} = 1$

$Din\ ecuatia\ (5) \Longrightarrow l_{11} = \frac{9}{2}$

$Din\ ecuatia\ (6) \Longrightarrow l_{10} = 3$

$Din\ ecuatia\ (7) \Longrightarrow \frac{9}{2}u_{01} = \frac{5}{2} \Longrightarrow u_{01} = \frac{5}{9}$

$Din\ ecuatia\ (8) \Longrightarrow u_{00}l_{00} + \frac{5}{9} * 3 = \frac{9}{5} \Longrightarrow u_{00}l_{00} = \frac{9}{5} - \frac{5}{3} = \frac{2}{15}$

$$Alegem\ u_{00} = 1 \Longrightarrow l_{00} = \frac{2}{15}$$

$$Am\ obtinut:$$

$$U = \begin{pmatrix} 1 & \frac{5}{9} & \frac{1}{10} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix} \qquad L = \begin{pmatrix} \frac{2}{15} & 0 & 0 \\ 3 & \frac{9}{2} & 0 \\ 2 & 5 & 10 \end{pmatrix}$$

In general, matricele L respectiv U din factorizarile LU sau UL ***difera****. Factorizarile furnizeaza acelesi matrice L respectiv U doar daca acestea* ***comuta*** *(se intampla rar).*

Exerciții propuse:

$Sa\ se\ rezolve\ sistemele\ Ax = b\ date\ de\ factorizarile\ matricelor\ din\ Tutoriatele\ 1\ si\ 2$

$Indicatii$:

$$Ax = B$$

$$(LU)x = b \implies L(Ux) = b. Notam\ Ux \to y\ si\ rezolvam\ intai\ Ly = b\ (aflam\ y)$$

$$Stim\ ca\ Ux = y\ (acum\ stiut), deci\ calculam\ x\ Q.E.D.$$

Matrice de factorizat:

$$A = \begin{pmatrix} 8 & 7 & 4 \\ 7 & 9 & 3 \\ 12 & 12 & 12 \end{pmatrix}, B = \begin{pmatrix} 3 & 8 & 1 \\ -2 & 9 & 3 \\ -1 & 2 & 1 \end{pmatrix}, C = \begin{pmatrix} -3 & 4 & 6 \\ 4 & -3 & -4 \\ -4 & 4 & 8 \end{pmatrix}$$